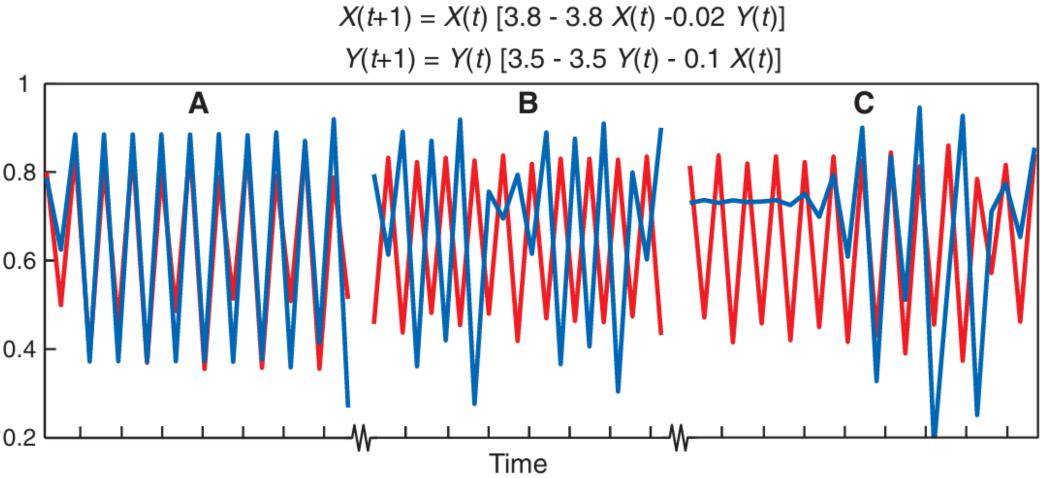
Causal discovery in time series data with empirical dynamic modeling

The R package tEDM

Wenbo Lyu, Wufan Zhao*

Urban Governance and Design Thrust, Society Hub
The Hong Kong University of Science and Technology (Guangzhou)

April 25, 2026



A, B and C: positive, negative, and zero correlation; Totally, n = 1000, r = 0.0054, P = 0.864

The Difficulty and Importance of Causal Discovery

From Data to Decisions: Finding the True Cause, Not Just Correlation

Why It Matters

Make Better Decisions
Identify what truly drives outcomes to act effectively.

Avoid Harm
Avoid interventions on variables that only look important.

Generalize to New Settings
Causal knowledge transfers beyond the data we observed.

Understand Mechanisms
Reveal how and why changes produce effects.

Correlation Is Not Causation

Correlation (Misleading)

X ← C → Y

X and Y are associated because of C, not because X causes Y.

Causation (What We Want)

X → Y

X has a (possibly direct or indirect) causal effect on Y.

Many Equivalent Stories Fit the Same Data

Direct: X → Y

Confounded: X ← C → Y

Mediated: X → M → Y

Collider: X → C ← Y

Different graphs can explain the data equally well, but imply different actions and conclusions.

Causal discovery is hard because the same data can be explained by many different causal structures.
We must use assumptions, domain knowledge, and better data.

Why It's Hard

Many Possibilities
Exponential number of graphs for even moderate variables.

Limited & Noisy Data
We observe only a fraction of the true data-generating process.

Hidden Confounders
Unmeasured causes can create spurious associations.

Causal Assumptions
We must assume things (e.g., no hidden confounding, correct interventions).

Context Dependence
Causal effects can change across populations and time.

How We Make Progress

Stronger Data
Experiments, natural experiments, longitudinal data.

+

Domain Knowledge
Theory, constraints, invariants, and prior information.

+

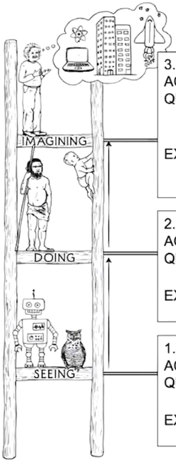
Better Algorithms
Structure learning, causal tests, and robust methods.

+

Better Decisions
Intervene on causes, not symptoms.

The Ladder of Causation

3-LEVEL HIERARCHY



3. COUNTERFACTUALS
ACTIVITY: Imagining, Retrospection, Understanding
QUESTIONS: *What if I had done . . . ? Why?*
(Was it X that caused Y? What if X had not occurred? What if I had acted differently?)
EXAMPLES: Was it the aspirin that stopped my headache?
Would Kennedy be alive if Oswald had not killed him? What if I had not smoked the last 2 years?

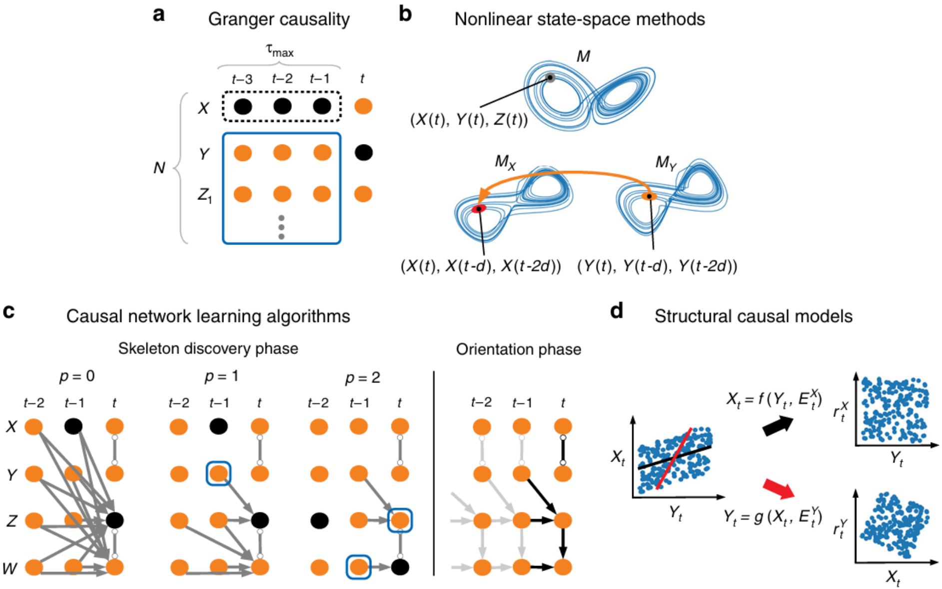
$$P(Y_{X'}|X)$$

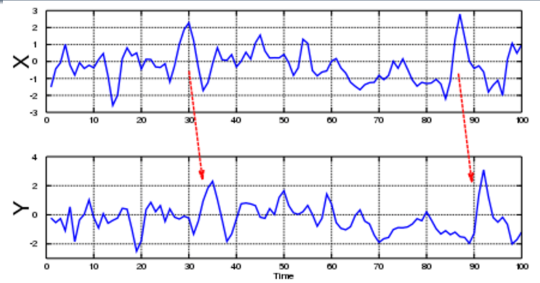
2. INTERVENTION
ACTIVITY: Doing, Intervening
QUESTIONS: *What if I do . . . ? How?*
(What would Y be if I do X?)
EXAMPLES: If I take aspirin, will my headache be cured?
What if we ban cigarettes?

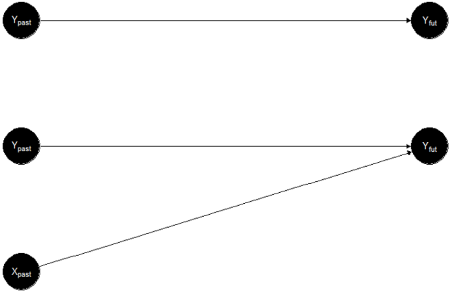
$$P(Y|do(X)), P(Y_x), P(Y(x))$$

1. ASSOCIATION
ACTIVITY: Seeing, Observing
QUESTIONS: *What if I see . . . ?*
(How would seeing X change my belief in Y?)
EXAMPLES: What does a symptom tell me about a disease?
What does a survey tell us about the election results?

$$P(Y|X)$$


a
Granger causality
τ_{max}
t–3 t–2 t–1 t
X
Y
Z_1
N
b
Nonlinear state-space methods
M
(X(t), Y(t), Z(t))
M_X
M_Y
(X(t), X(t-d), X(t-2d))
(Y(t), Y(t-d), Y(t-2d))
c
Causal network learning algorithms
Skeleton discovery phase
Orientation phase
p = 0
p = 1
p = 2
t–2 t–1 t
X
Y
Z
W
d
Structural causal models
$X_t = f(Y_t, E^X_t)$
$Y_t = g(X_t, E^Y_t)$
X_t
Y_t
r^X_t
r^Y_t





Granger Causality

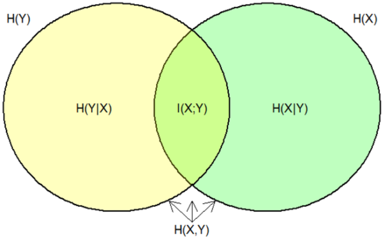
$$Y(t) = \sum_{j=1}^{p} \alpha_j Y(t-j) + \xi_1(t) \tag{1}$$

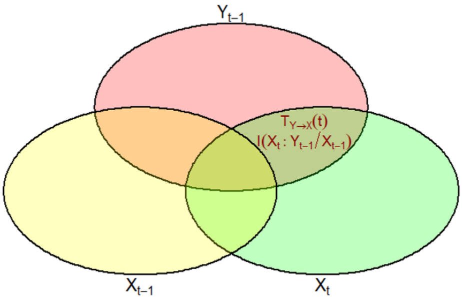
$$Y(t) = \sum_{j=1}^{p} \alpha_j Y(t-j) + \sum_{j=1}^{p} \beta_j X(t-j) + \xi_2(t) \tag{2}$$

A F-test is performed with *the null hypothesis of $Y(t)$ equals to model* (1) against *the alternative hypothesis of $Y(t)$ equals to model* (2).

We say that X Granger causes Y if we reject the null hypothesis.

Transfer Entropy

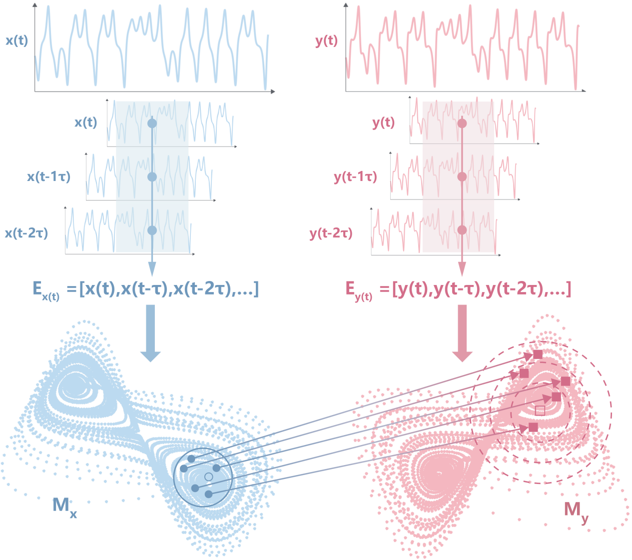




$$H(X) = -\sum_{x \in X} p(x) \log_2 p(x)$$

$$\begin{aligned} T_{Y\to X}(t) &= I(X_t : Y_{t-1} | X_{t-1}) \\ &= H(X_t|X_{t-1}) - H(X_t|X_{t-1}\, Y'_{t-1}) \\ &= \sum p(x_t, x_{t-1}, y_{t-1}) * \log_2 \frac{p(x_t|x_{t-1} y'_{t-1})}{p(x_t|x_{t-1})} \end{aligned}$$

- Transfer entropy can be considered a **non-parametric equivalent** of Granger Causality (it also works for *nonlinear categorical variables*).
- The **mutual information** between both is **symmetric** (undirected), but the introduced **time delay** allows for establishing directionality.



Empirical Dynamic Modeling (EDM)

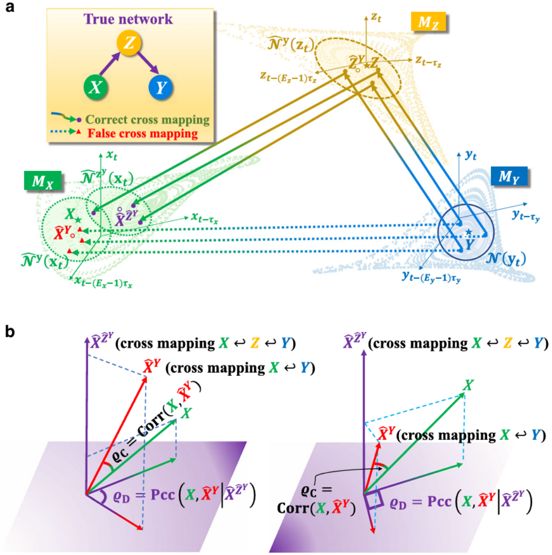
Time-delay Embedding

$$M_x = \left[x(t), x(t-\tau), \ldots, x\big(t-(E-1)\tau\big)\right]$$
$$M_y = \left[y(t), y(t-\tau), \ldots, y\big(t-(E-1)\tau\big)\right]$$

Mutual Cross Mapping

$$\hat{X}_t \mid M_y = \sum_{i=1}^{k} \left(\omega_{ti} X_{ti} \mid M_y\right)$$

$$\rho = \frac{\mathrm{Cov}\left(X, \hat{X} \mid M_y\right)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}\left(\hat{X} \mid M_y\right)}}$$



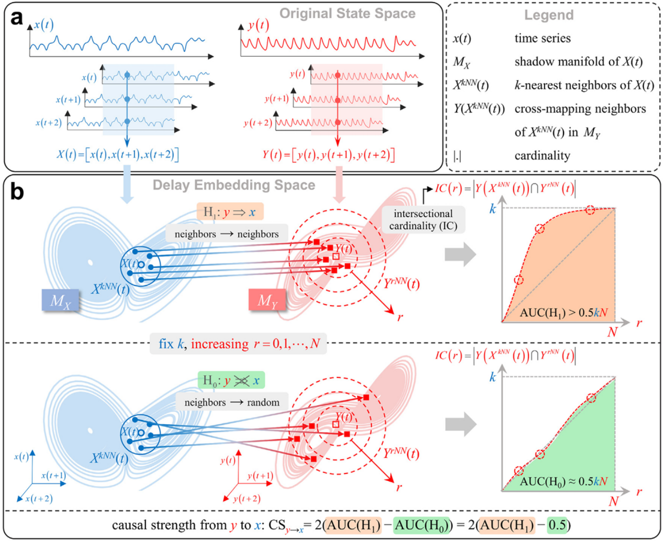
Partial Cross Mapping (PCM)

Accounting for indirect causal influences mediated by third variables

$$\hat{Z}_t \mid M_y = \sum_{i=1}^{k} \left(\omega_{ti} Z_{ti} \mid M_y\right)$$

$$\hat{X}_t^{(Z)} \mid M_{\hat{Z}} = \sum_{i=1}^{k} \left(\omega_{ti}^{(Z)} X_{ti} \mid M_{\hat{Z}}\right)$$

$$\rho_{X \to Y|Z} = \frac{\rho_{X,\hat{X}} - \rho_{X,\hat{X}^{(Z)}} \rho_{\hat{X},\hat{X}^{(Z)}}}{\sqrt{\left(1 - \rho^2_{X,\hat{X}^{(Z)}}\right)\left(1 - \rho^2_{\hat{X},\hat{X}^{(Z)}}\right)}}$$



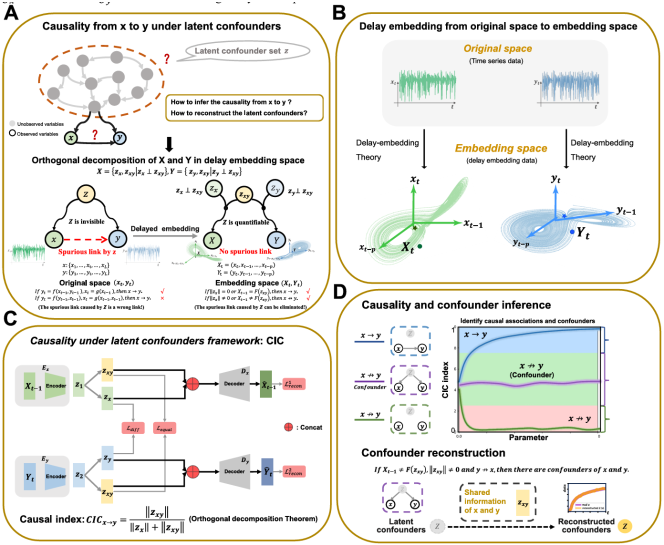
Cross Mapping Cardinality (CMC)

Employing neighborhood topological consistency in place of direct prediction

$$IC_{i,k} = \left| M_y^{nn}(i,k) \cap M_y^{nn}\left(M_x^{nn}(i,k),k\right) \right|$$

$$IC_k = \frac{1}{T}\sum_{i=1}^{T} IC_{i,k}$$

$$IC_{\text{curve}} = \{IC_k \mid k = 1,2,\ldots,T\}$$



Dynamical Causality under Latent Confounders (CIC)

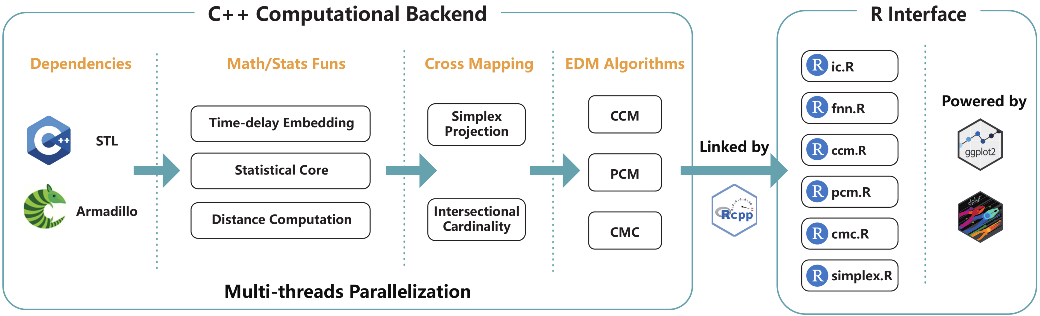
- Mathematically separates coupled dynamics into orthogonal common (confounder) and private subspaces.
- Accurately infers causal directions using **only two observed variables**

Open-source toolkits for causal discovery and their applicability to time series.

Package	Language	Implemented methods	Time series ready
dowhy-gcm	Python	Graphical causal modeling	No
causal-learn	Python	Multiple causal discovery algorithms	Partial
cdt	Python	Causal discovery toolbox (hybrid methods)	Partial
lingam	Python	Linear non-Gaussian models	Yes
tigramite	Python	Causal time series analysis via PCMCI and its variants	Yes
pcalg	R	Causal structure learning	Partial
bnlearn	R	Bayesian network structure learning	Partial
lmtest	R	Granger causality tests (GCT)	Yes
NliTS	R	Nonlinear time series causality methods	Yes
rEDM & *pyEDM*	R & Python	Convergent Cross Mapping (CCM)	Yes
multispatialccm	R	Multispatial Convergent Cross Mapping (MultispatialCCM)	Yes
fastEDM	R & Python	CCM & MultispatialCCM	Yes
crossmapy	Python	Temporal Empirical Dynamic Modeling	Yes
tEDM (Our work)	R	Enhanced Temporal Empirical Dynamic Modeling	Yes

tEDM integrates the state-of-the-art EDM causal discovery algorithms, with enhanced spatial relationship representations.

Overview of the source code architecture of the tEDM package



tEDM

Temporal Empirical Dynamic Modeling

Overview

The `tEDM` package provides a suite of tools for exploring and quantifying causality in time series using Empirical Dynamic Modeling (EDM). It is particularly designed to detect, differentiate, and reconstruct causal dynamics in systems where traditional assumptions of linearity and stationarity may not hold.

The package implements four fundamental EDM-based methods:

- Convergent Cross Mapping (CCM) – for detecting nonlinear causal relationships in time series.
- Partial Cross Mapping (PCM) – for disentangling direct from indirect causal influences.
- Cross Mapping Cardinality (CMC) – for identifying time-varying or state-dependent causal linkages.
- Multispatial Convergent Cross Mapping (MultispatialCCM) – for reconstructing causal dynamics from replicated time series across multiple spatial locations.

> *Refer to the package documentation https://stscl.github.io/tEDM/ for more detailed information.*

Installation

- Install from CRAN with:

```
install.packages("tEDM", dep = TRUE)
```

Links

View on CRAN

Browse source code

Report a bug

License

GPL-3

Citation

Citing tEDM

Developers

Wenbo Lyu

Author, maintainer, copyright holder

Dev status

CRAN 1.3

CRAN 2026-03-30

CRAN NOTE

downloads 2.8K

downloads 575/month

license GPL-3

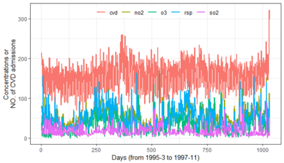
R-CMD-check passing

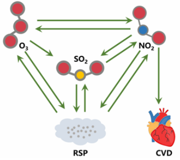
lifecycle stable

r-universe 2.0

CEUS 10.1016/j.compenvurbsys.2026.102435

DOI 10.5281/zenodo.18252239





Thank you for listening !

lyu.geosocial@gmail.com

https://github.com/SpatLyu/tEDM_Slide

https://cran.r-project.org/package=tEDM